MONOTONICITY OF ERDŐS SUMS

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ABSTRACT. For $k \geq 1$ let $f_k = \sum_{\Omega(n)=k} 1/(n \log n)$, where Ω counts prime factors with multiplicity. We prove that $f_k < f_{k+1}$ for every $k \geq 6$, whereas the sums restricted to odd integers satisfy $g_k > g_{k+1}$ for every $k \geq 1$. The two statements arise from the same analytic distinction. After writing the relevant Dirichlet series as $\zeta(s)^z H(s,z)$, the point $z = 2$ lies on the first singular circle for the unrestricted product and strictly inside that circle after the prime 2 is removed. The analytic reductions are written explicitly below. The finite inequalities and coefficient bounds used in the last step are certified by ball arithmetic; the separation between analytic argument and certified input is stated precisely.

1. INTRODUCTION

For $k \geq 1$ put

$$f_k = \sum_{\Omega(n)=k} \frac{1}{n \log n}.$$

Erdős [4] proved that the corresponding sum is bounded uniformly over primitive sets. In particular each f_k converges. Zhang [7] proved $f_k < f_1$ for $k > 1$, and Bayless, Kinlaw and Klyve [8] proved $f_2 > f_3$. Banks and Martin [1] conjectured that

$$f_1 > f_2 > f_3 > \cdots . \tag{1}$$

Lichtman [2] disproved (1), showing that $f_k \to 1$ and that the sequence attains a global minimum at $k = 6$. Gorodetsky, Lichtman and Wong [3] then established

$$f_k = 1 - 2^{-k}\big(a_1 k^2 + O(k \log(k+1))\big), \qquad a_1 = \frac{d \log 2}{4} > 0, \tag{2}$$

where

$$d = \frac{1}{4} \prod_{p \geq 3} \left(1 - \frac{2}{p}\right)^{-1} \left(1 - \frac{1}{p}\right)^{2}. \tag{3}$$

The implicit constant in (2) does not locate the first index at which the sequence begins to increase. Our first theorem does.

Theorem 1.1. *For every $k \geq 6$ one has $f_k < f_{k+1}$.*

Let

$$g_k = \sum_{\substack{n \text{ odd} \\ \Omega(n)=k}} \frac{1}{n \log n}.$$

The second theorem shows that deleting the prime 2 reverses the direction of monotonicity at every index.

Theorem 1.2. *For every $k \geq 1$ one has $g_k > g_{k+1}$, and $g_k \to \frac{1}{2}$.*

The proof of Theorem 1.2 gives the more precise expansion

$$g_k = \frac{1}{2} + \beta 2^{-k} + R_k, \qquad |R_k| \le 4900(k+3)^3 3^{-k}, \tag{4}$$

where

$$\beta = d\left(\gamma + \log 2 - \sum_{p\ge 3} \frac{\log p}{(p-1)(p-2)}\right) > 0. \tag{5}$$

The 2^{-k} term comes from evaluating a regularized Euler product at $z = 2$. For odd integers the first pole is at $z = 3$, so this evaluation takes place strictly inside the disc of analyticity. For the unrestricted sums the first pole is at $z = 2$ itself. The boundary contribution then acquires the factor k^2 in (2).

The proof has two kinds of input. Identities, coefficient reductions and inequalities are proved in the text. Bounds produced by interval arithmetic are isolated as certified propositions. Thus a sentence of the form "the computation gives" never substitutes for an analytic step: the exact inequality supplied by the computation is stated before it is used.

2. Generating functions and the analytic boundary

For $s > 1$ define

$$P_k(s) = \sum_{\Omega(n)=k} n^{-s}, \qquad P_k^{\mathrm{odd}}(s) = \sum_{\substack{n \text{ odd} \\ \Omega(n)=k}} n^{-s}.$$

Lemma 2.1. *For every $k \ge 1$,*

$$f_k = \int_1^\infty P_k(s)\,\mathrm{d}s, \qquad g_k = \int_1^\infty P_k^{\mathrm{odd}}(s)\,\mathrm{d}s. \tag{6}$$

Proof. For $n \ge 2$,

$$\int_1^\infty n^{-s}\,\mathrm{d}s = \int_1^\infty \mathrm{e}^{-s\log n}\,\mathrm{d}s = \left[-\frac{\mathrm{e}^{-s\log n}}{\log n}\right]_{s=1}^{\infty} = \frac{1}{n\log n}.$$

All summands are non-negative. Tonelli's theorem therefore permits the sum and integral to be interchanged in both cases, giving

$$\int_1^\infty \sum_{\Omega(n)=k} n^{-s}\,\mathrm{d}s = \sum_{\Omega(n)=k} \int_1^\infty n^{-s}\,\mathrm{d}s = f_k,$$

and the same calculation with the sum restricted to odd n proves the second identity. □

The generating functions are

$$\sum_{k\ge 0} P_k(s) z^k = \prod_p (1 - zp^{-s})^{-1}, \qquad \sum_{k\ge 0} P_k^{\mathrm{odd}}(s) z^k = \prod_{p\ge 3} (1 - zp^{-s})^{-1}. \tag{7}$$

Write these products as $\zeta(s)^z H(s,z)$ and $\zeta(s)^z H_{\mathrm{o}}(s,z)$, where

$$H(s,z) = \prod_p (1 - zp^{-s})^{-1}(1 - p^{-s})^z, \tag{8}$$

$$H_{\mathrm{o}}(s,z) = (1 - 2^{-s})^z \prod_{p\ge 3} (1 - zp^{-s})^{-1}(1 - p^{-s})^z. \tag{9}$$

Proposition 2.2. *For fixed real $s > 1/2$, the function $H(s,\cdot)$ is analytic in $|z| < 2^s$, and $H_{\mathrm{o}}(s,\cdot)$ is analytic in $|z| < 3^s$. The convergence is moreover locally uniform in (s,z) jointly, so each $\partial_s^r H_{\mathrm{o}}(s,\cdot)$ is again analytic in $|z| < 3^s$, and*

$$\frac{\mathrm{d}^r}{\mathrm{d}s^r}[z^j]H_{\mathrm{o}}(s,z) = [z^j]\partial_s^r H_{\mathrm{o}}(s,z); \tag{10}$$

the same holds for H in $|z| < 2^s$.

Proof. Let $x = p^{-s}$. For z in a fixed compact set, the logarithm of the pth regularized factor is

$$\begin{aligned} -\log(1-zx) + z\log(1-x) &= \sum_{m\ge 1} \frac{z^m - z}{m} x^m \\ &= \sum_{m\ge 2} \frac{z^m - z}{m} x^m = O_K(p^{-2s}). \end{aligned}$$

The estimate is uniform for z in the compact set K. Since $\sum_p p^{-2s} < \infty$, the product of the regularized factors converges locally uniformly wherever none of the factors has a pole. The poles occur at $z = p^s$. The nearest one is 2^s for (8) and 3^s for (9). A locally uniform limit of analytic functions is analytic, which proves the first assertion.

The estimate is uniform in s as well on any compact subset of $(1/2,\infty)$, since $p^{-2s} \le p^{-2s_0}$ there, and on a compact subset of $\{(s,z) : |z| < 3^s\}$ no factor comes near a pole. The convergence is therefore locally uniform in the two variables jointly, so differentiation in s may be carried out under the product. Since $[z^j]$ is a Cauchy integral over a circle that may be held fixed on such a subset, the two operations commute, which is (10). □

At $s = 1$ the point $z = 2$ lies on the first singular circle for H and lies strictly inside the first singular circle for H_{o}. This is the analytic distinction used below.

We shall also need the local change of variables used in the odd case.

Lemma 2.3. *Let $u = \log\zeta(s)$ for $s > 1$, and let $s = s(u)$ be the inverse function. Put $\varepsilon(u) = s(u) - 1$ and $J(u) = -s'(u)$. As $u \to \infty$,*

$$\varepsilon(u) = \mathrm{e}^{-u} + \gamma\mathrm{e}^{-2u} + O(\mathrm{e}^{-3u}), \qquad J(u) = \mathrm{e}^{-u} + 2\gamma\mathrm{e}^{-2u} + O(\mathrm{e}^{-3u}). \tag{11}$$

Proof. For $\varepsilon \to 0^+$,

$$\zeta(1+\varepsilon) = \varepsilon^{-1} + \gamma + O(\varepsilon).$$

Set $x = \mathrm{e}^{-u} = 1/\zeta(1+\varepsilon)$. Then

$$x = \frac{\varepsilon}{1 + \gamma\varepsilon + O(\varepsilon^2)} = \varepsilon - \gamma\varepsilon^2 + O(\varepsilon^3).$$

Inverting this power series gives $\varepsilon = x + \gamma x^2 + O(x^3)$, which is the first formula in (11). Since $x' = -x$, differentiation gives

$$\varepsilon'(u) = -x - 2\gamma x^2 + O(x^3).$$

Thus $J(u) = -\varepsilon'(u) = x + 2\gamma x^2 + O(x^3)$. □

3. The odd sums

Throughout this section write

$$h_j(s) = [z^j]H_{\mathrm{o}}(s,z), \qquad b_j = h_j'(1) + 2\gamma h_j(1).$$

Since $\zeta(s)^z = \sum_{m\geq 0}(\log\zeta(s))^m z^m/m!$, coefficient extraction in (7) gives the exact identity

$$P_k^{\mathrm{odd}}(s) = \sum_{j=0}^{k} h_j(s)\frac{(\log\zeta(s))^{k-j}}{(k-j)!}. \tag{12}$$

Using $u = \log\zeta(s)$ in Lemma 2.1, and noting that s decreases from ∞ to 1 as u increases from 0 to ∞, we obtain

$$g_k = \sum_{j=0}^{k}\int_0^\infty \frac{u^{k-j}}{(k-j)!}\, h_j(s(u))J(u)\,\mathrm{d}u. \tag{13}$$

3.1. **The first two terms.**

Proposition 3.1. *There is a remainder R_k such that*

$$g_k = H_{\mathrm{o}}(1,1) + 2^{-k-1}\big(\partial_s H_{\mathrm{o}}(1,2) + 2\gamma H_{\mathrm{o}}(1,2)\big) + R_k. \tag{14}$$

Proof. Let $x = \mathrm{e}^{-u}$. By Lemma 2.3, $J(u) = x + 2\gamma x^2 + O(x^3)$ and $\varepsilon(u)J(u) = x^2 + O(x^3)$. Taylor's theorem at $s = 1$ gives

$$h_j(1+\varepsilon) = h_j(1) + \varepsilon h_j'(1) + \frac{\varepsilon^2}{2}h_j''(1+\theta_{j,u}\varepsilon)$$

for some $0 < \theta_{j,u} < 1$. Put

$$r_0(u) = J(u) - \mathrm{e}^{-u} - 2\gamma\mathrm{e}^{-2u}, \qquad r_1(u) = \varepsilon(u)J(u) - \mathrm{e}^{-2u}.$$

Multiplication by $J(u)$ now gives the exact identity

$$\begin{aligned} h_j(s(u))J(u) &= h_j(1)\mathrm{e}^{-u} + b_j\mathrm{e}^{-2u} + E_j(u),\\ E_j(u) &= h_j(1)r_0(u) + h_j'(1)r_1(u) + \frac{\varepsilon(u)^2 J(u)}{2}h_j''(1+\theta_{j,u}\varepsilon(u)). \end{aligned} \tag{15}$$

Thus no error is suppressed in the notation E_j: its three summands are the Jacobian error, the first-order coordinate error and the Taylor remainder.

Insert (15) into (13). Since

$$\int_0^\infty \frac{u^m}{m!}\mathrm{e}^{-ru}\,\mathrm{d}u = r^{-m-1} \qquad (r>0),$$

the first two terms become

$$\sum_{j=0}^{k} h_j(1) = H_{\mathrm{o}}(1,1) - \sum_{j>k} h_j(1), \tag{16}$$

$$\sum_{j=0}^{k} b_j 2^{-(k-j+1)} = 2^{-k-1}\big(\partial_s H_{\mathrm{o}}(1,2) + 2\gamma H_{\mathrm{o}}(1,2)\big) - 2^{-k-1}\sum_{j>k} b_j 2^j. \tag{17}$$

Define R_k to be the integral of the E_j terms together with the two tails on the right of (16) and (17). This gives (14). □

Lemma 3.2. *One has*

$$H_{\mathrm{o}}(1,1) = \frac{1}{2}, \qquad H_{\mathrm{o}}(1,2) = d, \tag{18}$$

and

$$\partial_s \log H_{\mathrm{o}}(1,2) = 2\log 2 - 2\sum_{p\geq 3}\frac{\log p}{(p-1)(p-2)}. \tag{19}$$

Proof. At $s = z = 1$, every factor in the product over $p \geq 3$ cancels, and hence

$$H_{\mathrm{o}}(1,1) = (1 - 2^{-1}) = \frac{1}{2}.$$

At $(s, z) = (1, 2)$,

$$H_{\mathrm{o}}(1,2) = \frac{1}{4} \prod_{p \geq 3} \left(1 - \frac{2}{p}\right)^{-1} \left(1 - \frac{1}{p}\right)^{2} = d.$$

For the derivative, take logarithms:

$$\log H_{\mathrm{o}}(s,2) = 2\log(1 - 2^{-s}) + \sum_{p \geq 3} \left[-\log(1 - 2p^{-s}) + 2\log(1 - p^{-s}) \right].$$

The differentiated pth summand is $O((\log p)p^{-2s})$ locally uniformly near $s = 1$, so termwise differentiation is permitted. At $s = 1$ the first term contributes $2\log 2$, while the pth summand contributes

$$-\frac{2\log p}{p-2} + \frac{2\log p}{p-1} = -\frac{2\log p}{(p-1)(p-2)}.$$

Summing over $p \geq 3$ proves (19). □

Combining Proposition 3.1 and Lemma 3.2 gives

$$g_k = \frac{1}{2} + \beta 2^{-k} + R_k, \qquad \beta = d\left(\gamma + \log 2 - \sum_{p \geq 3} \frac{\log p}{(p-1)(p-2)}\right). \tag{20}$$

3.2. **The remainder.** The following proposition records exactly the interval-arithmetic input used for the odd sums. The coefficient sums are weighted ℓ^1 norms; their finiteness follows analytically from Proposition 2.2, while the displayed numerical bounds are certified.

Proposition 3.3 (Certified odd input)**.** *Let*

$$\Sigma_0 = \sum_{m \geq 0} 3^m |[z^m](1 - z/3)H_{\mathrm{o}}(1,z)|,$$
$$\Sigma_1 = \sum_{m \geq 0} 3^m |[z^m](1 - z/3)^2 \partial_s H_{\mathrm{o}}(1,z)|,$$
$$\Sigma_2 = \sup_{1 \leq s \leq s(3)} \sum_{m \geq 0} 3^m |[z^m](1 - z/3)^3 \partial_s^2 H_{\mathrm{o}}(s,z)|.$$

The certified computation gives

$$\Sigma_0 \leq 44.5987, \qquad \Sigma_1 \leq 486.1945, \qquad \Sigma_2 \leq 14414. \tag{21}$$

Split the remainder of Proposition 3.1 at $u = 3$: with E_j the explicit expression in (15)*, put*

$$R_{k,\mathrm{far}} = \sum_{j=0}^{k} \int_3^\infty \frac{u^{k-j}}{(k-j)!} E_j(u)\,\mathrm{d}u, \qquad R_{k,\mathrm{near}} = \sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} E_j(u)\,\mathrm{d}u, \tag{22}$$

so that R_k is $R_{k,\mathrm{far}} + R_{k,\mathrm{near}}$ together with the two coefficient tails of (16)–(17)*. For the far range the Taylor-model computation certifies the aggregate inequality*

$$\sum_{j=0}^{k} \int_3^\infty \frac{u^{k-j}}{(k-j)!} |E_j(u)|\,\mathrm{d}u \leq \frac{C_E \Sigma_2}{18}(k+3)^3 3^{-k}, \qquad C_E \leq 5.48910. \tag{23}$$

For the near range the two groups of terms in (15) *are certified separately. With* $s(3)$ *the value of* s *at* $u = 3$*,*

$$P_k^{\mathrm{odd}}(s(3)) \le C_N 3^{-k} \quad (k \ge 1), \qquad C_N \le 367.8435, \tag{24}$$

and the two subtracted main terms satisfy

$$\sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} \big(|h_j(1)|\mathrm{e}^{-u} + |b_j|\mathrm{e}^{-2u}\big)\,\mathrm{d}u \le N_U (k+3)^3 3^{-k}, \qquad N_U \le 26.0962. \tag{25}$$

Finally, $\beta > 0$*, the inequality*

$$4900(62)^3 3^{-59} < \frac{\beta}{3} 2^{-59} \tag{26}$$

is certified, and direct interval evaluation gives $g_k > g_{k+1}$ *for* $1 \le k \le 75$*.*

Lemma 3.4. *Let* $A(z) = \sum_{j\ge 0} a_j z^j$*.*
(i) If $\sum_m 3^m |[z^m](1 - z/3)A(z)| \le S$*, then* $|a_j| \le S3^{-j}$*.*
(ii) If $\sum_m 3^m |[z^m](1 - z/3)^2 A(z)| \le S$*, then* $|a_j| \le S(j+1)3^{-j}$*.*
(iii) If $\sum_m 3^m |[z^m](1 - z/3)^3 A(z)| \le S$*, then*

$$\sum_{j=0}^{k} 3^j |a_j| \le \binom{k+3}{3} S. \tag{27}$$

Proof. Write $(1 - z/3)^r A(z) = \sum_m w_m z^m$. Then

$$A(z) = \left(\sum_{q\ge 0} \binom{q+r-1}{r-1} 3^{-q} z^q \right) \left(\sum_{m\ge 0} w_m z^m \right).$$

For $r = 1$ this gives $|a_j| \le 3^{-j} \sum_{m\le j} 3^m |w_m| \le S3^{-j}$. For $r = 2$ it gives $|a_j| \le (j+1)3^{-j}S$. For $r = 3$,

$$\begin{aligned} \sum_{j=0}^{k} 3^j |a_j| &\le \sum_{m=0}^{k} 3^m |w_m| \sum_{q=0}^{k-m} \binom{q+2}{2} \\ &= \sum_{m=0}^{k} 3^m |w_m| \binom{k-m+3}{3} \le \binom{k+3}{3} S. \end{aligned}$$

□

Proposition 3.5. *For every* $k \ge 1$*,*

$$|R_k| \le 4900(k+3)^3 3^{-k}. \tag{28}$$

Proof. We bound separately the far Taylor remainder, the two coefficient tails in (16)–(17), and the near range $0 < u \le 3$.

For the far range $u \ge 3$, the exact error is the sum of the three terms in (15); no asymptotic series is summed. The certified Taylor model gives directly

$$|R_{k,\mathrm{far}}| \le \frac{C_E \Sigma_2}{18} (k+3)^3 3^{-k} \tag{29}$$

by (23).

Lemma 3.4(i) gives

$$|\sum_{j>k} h_j(1)| \le \Sigma_0 \sum_{j>k} 3^{-j} = \frac{\Sigma_0}{2} 3^{-k}. \tag{30}$$

For the second tail, first put $B(z) = \partial_s H_{\mathrm{o}}(1,z) + 2\gamma H_{\mathrm{o}}(1,z)$. Since $\gamma < 0.578$, multiplication by $1 - z/3$ and the triangle inequality give

$$\sum_{m\ge 0} 3^m |[z^m](1-z/3)^2 B(z)| \le \Sigma_1 + 4\gamma\Sigma_0 < 590.$$

Hence Lemma 3.4(ii) gives $|b_j| \le 590(j+1)3^{-j}$. Therefore

$$\begin{aligned} 2^{-k-1}\sum_{j>k} |b_j| 2^j &\le 590\, 2^{-k-1} \sum_{j>k} (j+1)(2/3)^j \\ &= 590(k+4)3^{-k} \\ &\le \frac{5\cdot 590}{64}(k+3)^3 3^{-k}, \end{aligned} \tag{31}$$

where the last inequality follows from $(k+4)/(k+3)^3 \le 5/64$ for $k \ge 1$.

For the near range, $|E_j|$ is at most the true integrand plus the two subtracted main terms. The latter contribute at most $N_U(k+3)^3 3^{-k}$ by (25). For the former, (12) and the substitution $s = s(u)$ give the exact identity

$$\sum_{j=0}^{k} \int_0^3 \frac{u^{k-j}}{(k-j)!} h_j(s(u)) J(u)\, \mathrm{d}u = \int_{s(3)}^{\infty} P_k^{\mathrm{odd}}(s)\, \mathrm{d}s.$$

Every odd n with $\Omega(n) = k$ satisfies $n \ge 3^k$, so for $s \ge s(3)$

$$P_k^{\mathrm{odd}}(s) = \sum_{\substack{n \text{ odd} \\ \Omega(n)=k}} n^{-s(3)} n^{-(s-s(3))} \le 3^{-k(s-s(3))} P_k^{\mathrm{odd}}(s(3)),$$

and integrating over $s \ge s(3)$ gives, by (24),

$$\int_{s(3)}^{\infty} P_k^{\mathrm{odd}}(s)\, \mathrm{d}s \le \frac{P_k^{\mathrm{odd}}(s(3))}{k \log 3} \le \frac{C_N}{k\log 3} 3^{-k} \le C_N (k+3)^3 3^{-k}. \tag{32}$$

Hence $|R_{k,\mathrm{near}}| \le (C_N + N_U)(k+3)^3 3^{-k}$. Combining (29), (30), (31) and this bound, and then substituting the certified constants, gives

$$\begin{aligned} |R_k| &\le \left(\frac{5.48910 \cdot 14414}{18} + \frac{44.5987}{2} + \frac{5\cdot 590}{64} + 367.8435 + 26.0962 \right) (k+3)^3 3^{-k} \\ &< 4900(k+3)^3 3^{-k}. \end{aligned}$$

□

Proof of Theorem 1.2. The limit follows at once from (20) and Proposition 3.5. For monotonicity,

$$g_k - g_{k+1} = \beta 2^{-k-1} + R_k - R_{k+1}. \tag{33}$$

It is enough to prove $|R_m| < \beta 2^{-m}/3$ for $m = k, k+1$, because then

$$|R_k| + |R_{k+1}| < \frac{2\beta}{3} 2^{-k-1} + \frac{\beta}{3} 2^{-k-1} = \beta 2^{-k-1}.$$

Define

$$q(k) = \frac{3 \cdot 4900}{\beta}(k+3)^3(2/3)^k.$$

The certified inequality (26) says $q(59) < 1$. Moreover

$$\frac{q(k+1)}{q(k)} = \frac{2}{3}\left(\frac{k+4}{k+3}\right)^3 < 1 \qquad (k \geq 4).$$

Thus $q(k) < 1$ for every $k \geq 59$, and Proposition 3.5 gives $|R_k| < \beta 2^{-k}/3$ throughout that range. Equation (33) proves monotonicity for $k \geq 59$. The certified direct computation covers $1 \leq k \leq 75$, so the two ranges cover every $k \geq 1$. They overlap on sixteen indices. □

4. The unrestricted sums: an exact criterion

Write

$$f_k = 1 - 2^{-k}\big(a_1 k^2 + E(k)\big), \qquad a_1 = \frac{d \log 2}{4}. \tag{34}$$

Then

$$f_{k+1} - f_k = 2^{-k-1}\left(a_1(k^2 - 2k - 1) + 2E(k) - E(k+1)\right). \tag{35}$$

Thus $f_k < f_{k+1}$ if and only if

$$a_1(k^2 - 2k - 1) + 2E(k) - E(k+1) > 0. \tag{36}$$

We now derive the decomposition of $E(k)$ used to bound the second term.

Set

$$F(s,z) = \prod_p (1 - zp^{-s})^{-1}, \qquad G_2(s,z) = (s-1)^z(1 - z2^{-s})F(s,z),$$

and write $c_i = [z^i]G_2(1,z)$. Since $G_2(1,z) = H_{\mathrm{o}}(1,z)$, the power series has radius 3 and

$$\sum_{i \geq 0} c_i = \frac{1}{2}, \qquad \sum_{i \geq 0} 2^i c_i = d. \tag{37}$$

The coefficients are those of Section 3:

$$c_i = h_i(1). \tag{38}$$

Define the absolutely convergent signed moments

$$U = \sum_{i \geq 0} i 2^i c_i, \qquad V = \sum_{i \geq 0} i^2 2^i c_i. \tag{39}$$

Following [3, §3], put

$$I_k = \int_1^2 [z^k]F(s,z)\,\mathrm{d}s, \tag{40}$$

$$I_k' = \sum_{i+j+\ell=k} c_i \int_1^2 \frac{(-\log(s-1))^\ell}{\ell!} 2^{-js}\,\mathrm{d}s. \tag{41}$$

For $m \geq 0$ define

$$A_m = 2^m \sum_{j+\ell=m} \int_1^2 \frac{(-\log(s-1))^\ell}{\ell!} 2^{-js}\,\mathrm{d}s \tag{42}$$

and

$$\delta_m = A_m - 2^{m+1} + \frac{\log 2}{4} m^2, \qquad \Delta(k) = \sum_{i=0}^{k} 2^i c_i \delta_{k-i}. \tag{43}$$

Proposition 4.1. *Let*

$$E_{S_1}(k) = -2^k (I_k - I'_k), \tag{44}$$

$$L_k = 2 \sum_{i>k} c_i, \qquad Q_k = \sum_{i>k} 2^i c_i (k-i)^2, \tag{45}$$

$$T_k = f_k - I_k, \tag{46}$$

$$E_{\exp}(k) = 2^k L_k - \frac{\log 2}{4} Q_k - 2^k T_k. \tag{47}$$

Then the identity

$$E(k) = E_{S_1}(k) + \frac{\log 2}{4}(V - 2kU) - \Delta(k) + E_{\exp}(k) \tag{48}$$

holds for every $k \geq 1$.

Proof. From (41) and (42),

$$I'_k = 2^{-k} \sum_{i=0}^{k} 2^i c_i A_{k-i}.$$

Insert the definition of δ_{k-i}:

$$I'_k = 2 \sum_{i=0}^{k} c_i - \frac{\log 2}{4} 2^{-k} \sum_{i=0}^{k} 2^i c_i (k-i)^2 + 2^{-k} \Delta(k). \tag{49}$$

By (37) and (39),

$$2 \sum_{i=0}^{k} c_i = 1 - L_k,$$

$$\begin{aligned} \sum_{i=0}^{k} 2^i c_i (k-i)^2 &= \sum_{i \geq 0} 2^i c_i (k-i)^2 - Q_k \\ &= dk^2 - 2kU + V - Q_k. \end{aligned}$$

Substitution into (49) gives

$$I'_k = 1 - 2^{-k} \left(a_1 k^2 + \frac{\log 2}{4}(V - 2kU) - \Delta(k) + 2^k L_k - \frac{\log 2}{4} Q_k \right).$$

Finally $f_k = I'_k + (I_k - I'_k) + T_k$. Comparing with (34) yields (48). □

5. Bounds for the four source terms

5.1. **The S_1 term.** For $s \in [1, 2]$ set

$$\alpha_i = \sup_{1 \leq t \leq 2} |[z^i] \partial_s G_2(t, z)|, \qquad \alpha_+ = \frac{1}{2} \sum_{i \geq 0} \alpha_i 2^i.$$

The series converges because the first z-pole of $\partial_s G_2$ is at $z = 3^s$.

Lemma 5.1. *For every $k \geq 1$,*

$$|E_{S_1}(k)| \leq \alpha_+(k+1), \tag{50}$$

and hence

$$|2E_{S_1}(k) - E_{S_1}(k+1)| \leq \alpha_+(3k+4). \tag{51}$$

Proof. The mean value theorem gives

$$|[z^i]G_2(s,z) - c_i| \leq (s-1)\alpha_i.$$

Subtracting (41) from the coefficient expansion for I_k and then setting $s = 1 + \mathrm{e}^{-t}$ gives

$$\begin{aligned} |I_k - I_k'| &\leq \sum_{i+j+\ell=k} \alpha_i \int_0^\infty \frac{t^\ell}{\ell!} 2^{-j(1+\mathrm{e}^{-t})} \mathrm{e}^{-2t}\,\mathrm{d}t \\ &\leq \sum_{i+j+\ell=k} \alpha_i 2^{-j} \int_0^\infty \frac{t^\ell}{\ell!} \mathrm{e}^{-2t}\,\mathrm{d}t \\ &= \sum_{i+j+\ell=k} \alpha_i 2^{-j-\ell-1}. \end{aligned}$$

Multiplying by 2^k and using $k = i + j + \ell$,

$$|E_{S_1}(k)| \leq \frac{1}{2}\sum_{i=0}^{k} \alpha_i 2^i (k-i+1) \leq \alpha_+(k+1).$$

The triangle inequality now gives

$$|2E_{S_1}(k) - E_{S_1}(k+1)| \leq 2\alpha_+(k+1) + \alpha_+(k+2) = \alpha_+(3k+4).$$

□

5.2. **The signed moment term.**

Lemma 5.2. *If $|U| \leq U_{\max}$ and $|V| \leq V_{\max}$, then for $k \geq 2$,*

$$\frac{\log 2}{4}\big(V + (2-2k)U\big) \geq -\frac{\log 2}{4}\big(V_{\max} + (2k-2)U_{\max}\big). \tag{52}$$

Proof. The coefficient of V is 1, while the coefficient $2-2k$ of U is negative. The minimum over the rectangle $[-U_{\max}, U_{\max}] \times [-V_{\max}, V_{\max}]$ is therefore attained at $U = U_{\max}$ and $V = -V_{\max}$, giving (52). □

Indeed,

$$2(V - 2kU) - \big(V - 2(k+1)U\big) = V + (2-2k)U. \tag{53}$$

Thus Lemma 5.2 is an exact optimization, not a termwise absolute-value estimate.

5.3. **The δ term.** The next lemma supplies the self-contained bound used for $\Delta(k)$.

Lemma 5.3. *For $j \geq 0$ and $u \geq 0$ put*

$$B_j(u) = 2^{-j\mathrm{e}^{-u}} - 1 + j\log 2\,\mathrm{e}^{-u}$$

and

$$\eta_i(j) = \frac{1}{i!}\int_0^\infty u^i \mathrm{e}^{-u} B_j(u)\,\mathrm{d}u.$$

Then $B_j(u) \ge 0$, $\eta_i(j) \ge 0$, and

$$\delta_m = -1 - \frac{\log 2}{4} m + \sum_{i=0}^{m} 2^i \eta_i(m-i). \tag{54}$$

Proof. With $s = 1 + \mathrm{e}^{-u}$ in (42),

$$A_m = \sum_{i=0}^{m} \frac{2^i}{i!} \int_0^\infty u^i \mathrm{e}^{-u} 2^{-(m-i)\mathrm{e}^{-u}} \, \mathrm{d}u.$$

Use $2^{-j\mathrm{e}^{-u}} = 1 - j \log 2 \, \mathrm{e}^{-u} + B_j(u)$. The constant term contributes

$$\sum_{i=0}^{m} 2^i = 2^{m+1} - 1.$$

The linear term contributes

$$\begin{aligned} -\log 2 \sum_{i=0}^{m} \frac{2^i (m-i)}{i!} \int_0^\infty u^i \mathrm{e}^{-2u} \, \mathrm{d}u &= -\frac{\log 2}{2} \sum_{i=0}^{m} (m-i) \\ &= -\frac{\log 2}{4} m(m+1). \end{aligned}$$

The remaining term is $\sum_{i=0}^{m} 2^i \eta_i(m-i)$. Subtracting 2^{m+1} and adding $(\log 2) m^2/4$ gives (54). Finally, for $x \ge 0$, $\mathrm{e}^{-x} \ge 1 - x$; applying this with $x = j \log 2 \, \mathrm{e}^{-u}$ proves $B_j(u) \ge 0$. □

Lemma 5.4. *For every $m \ge 0$,*

$$|\delta_m| \le D(m), \qquad D(m) = (\log 2) m \log(m+1) + 1. \tag{55}$$

Proof. The lower bound follows at once from (54):

$$\delta_m \ge -1 - \frac{\log 2}{4} m \ge -D(m) \qquad (m \ge 1),$$

because $\log(m+1) \ge \log 2 > 1/4$. For $m = 0$, $\delta_0 = -1 = -D(0)$.

For the upper bound, write $T_m = \sum_{i=0}^{m} 2^i \eta_i(m-i)$. If $x \ge 0$, then $\mathrm{e}^{-x} - 1 + x \le \min(x, x^2/2)$. Hence, with $x = (m-i) \log 2 \, \mathrm{e}^{-u}$,

$$T_m \le \int_0^\infty \mathrm{e}^{-u} \sum_{i=0}^{m} \frac{(2u)^i}{i!} \min\left((m-i) \log 2 \, \mathrm{e}^{-u}, \frac{(m-i)^2 (\log 2)^2 \mathrm{e}^{-2u}}{2} \right) \mathrm{d}u. \tag{56}$$

Assume $m \ge 2$ and put $U_m = \log((m+1)/3) \ge 0$. On $0 \le u \le U_m$ use the linear bound and $\sum_{i=0}^{m} (m-i)(2u)^i/i! \le m\mathrm{e}^{2u}$; on $u \ge U_m$ use the quadratic bound and $\sum_{i=0}^{m} (m-i)^2 (2u)^i/i! \le m^2 \mathrm{e}^{2u}$. Equation (56) gives

$$\begin{aligned} T_m &\le m \log 2 \int_0^{U_m} 1 \, \mathrm{d}u + \frac{m^2 (\log 2)^2}{2} \int_{U_m}^\infty \mathrm{e}^{-u} \, \mathrm{d}u \\ &= m \log 2 \log \frac{m+1}{3} + \frac{3m^2 (\log 2)^2}{2(m+1)} \\ &\le m \log 2 \log(m+1) + m \log 2 \left(-\log 3 + \frac{3}{2} \log 2 \right) \\ &\le m \log 2 \log(m+1), \end{aligned}$$

where the last inequality uses $3 > 2^{3/2}$. Thus

$$\delta_m = -1 - \frac{\log 2}{4} m + T_m \le (\log 2) m \log(m+1) + 1 = D(m).$$

For $m = 1$, $B_0 = 0$ and $0 \le B_1(u) \le (\log 2)^2 \mathrm{e}^{-2u}/2$, so

$$0 \le T_1 = \eta_0(1) \le \frac{(\log 2)^2}{2} \int_0^\infty \mathrm{e}^{-3u}\,\mathrm{d}u = \frac{(\log 2)^2}{6}.$$

Hence $-D(1) < -1 - \frac{1}{4}\log 2 \le \delta_1 < 0 < D(1)$, which completes the remaining case. □

Lemma 5.5. *If $S^* = \sum_{i\ge 0} 2^i|c_i|$, then*

$$|2\Delta(k) - \Delta(k+1)| \le S^*\big(2D(k) + D(k+1)\big). \tag{57}$$

Proof. Since D is increasing,

$$|\Delta(k)| \le \sum_{i=0}^{k} 2^i|c_i|D(k-i) \le S^* D(k).$$

Apply the triangle inequality to $2\Delta(k) - \Delta(k+1)$. □

5.4. **The exponentially small term.** The three terms in (47) are respectively a coefficient tail at $z = 1$, a weighted coefficient tail at $z = 2$, and the integral over $s \ge 2$. The following is the exact numerical input used for them.

Proposition 5.6 (Certified unrestricted input)**.** *The interval computations give*

$$\begin{aligned} d \ge 0.3786949, \qquad S^* \le 6.9763, \qquad \alpha_+ \le 15.425661, \\ |U| \le U_{\max} = 0.08436, \qquad |V| \le V_{\max} = 1.907462. \end{aligned} \tag{58}$$

For every $k \ge 900$,

$$|E_{\exp}(k)| \le \frac{1.45}{k} + 10^{-3}. \tag{59}$$

The direct certificates establish $f_k < f_{k+1}$ on

$$6 \le k \le 199, \qquad 78 \le k \le 899, \qquad 900 \le k \le 2439. \tag{60}$$

The coefficient tails in (59) are obtained only after the pole at $z = 3$ has been removed. Put

$$\widetilde{G}_2(z) = (1 - z/3)G_2(1, z) = \sum_{m\ge 0} w_m z^m.$$

The certified contour bound is $\sup_{|z|=4.2}|\widetilde{G}_2(z)| \le 101.245$. Hence $|w_m| \le 101.245\,4.2^{-m}$, and from $G_2(1, z) = (1 - z/3)^{-1}\widetilde{G}_2(z)$ we obtain

$$|c_i| \le 101.245\,3^{-i} \sum_{m=0}^{i} (3/4.2)^m < 354.36\,3^{-i}. \tag{61}$$

By (38) the c_i are the coefficients $h_i(1)$, so Lemma 3.4(i) sharpens (61) to $|c_i| \le \Sigma_0 3^{-i}$, dispensing with the contour bound; the weaker form suffices below. With $C = 354.36$,

$$\begin{aligned} 2^k|L_k| &\le C(2/3)^k, \\ \frac{\log 2}{4}|Q_k| &\le \frac{\log 2}{4} C(2/3)^k \sum_{r\ge 1} r^2(2/3)^r = \frac{30C\log 2}{4}(2/3)^k. \end{aligned}$$

Both terms are less than 10^{-3} in total for $k \ge 900$. The remaining certified tail inequality is $2^k|T_k| \le 1.45/k$. Substitution into (47) proves (59); no asymptotic O-term enters the final criterion. The first two terms are in fact below 10^{-3} from $k \ge 37$ on, so the threshold 900 comes from T_k alone.

6. Assembly of the unrestricted theorem

For $k \geq 900$ define

$$R(k) = \frac{\log 2}{4}\big((2k-2)U_{\max} + 2V_{\max} + 4U_{\max}\big) + S^*\big(2D(k) + D(k+1)\big) + \alpha_+(3k+4) + 3\left(\frac{1.45}{k} + 10^{-3}\right), \tag{62}$$

and

$$\varphi(k) = a_1(k^2 - 2k - 1) - R(k). \tag{63}$$

Proposition 6.1. *For $k \geq 900$,*

$$2E(k) - E(k+1) \geq -R(k). \tag{64}$$

Consequently $\varphi(k) > 0$ implies $f_k < f_{k+1}$.

Proof. Apply (48) at k and $k+1$. The S_1 contribution is bounded by (51). The signed moment contribution is

$$\frac{\log 2}{4}\big(V + (2-2k)U\big),$$

which is bounded below by (52). The first line of (62) is slightly larger than the resulting quantity, since

$$V_{\max} + (2k-2)U_{\max} \leq (2k-2)U_{\max} + 2V_{\max} + 4U_{\max}.$$

The Δ contribution is bounded by (57). Finally, (59) gives

$$|2E_{\exp}(k) - E_{\exp}(k+1)| \leq 2\left(\frac{1.45}{k} + 10^{-3}\right) + \left(\frac{1.45}{k+1} + 10^{-3}\right) \leq 3\left(\frac{1.45}{k} + 10^{-3}\right).$$

Adding the four lower bounds proves (64). Equation (35) then gives

$$f_{k+1} - f_k \geq 2^{-k-1}\varphi(k),$$

which proves the last assertion. □

Lemma 6.2. *With the certified constants in Proposition 5.6, $\varphi(2440) > 1158$.*

Proof. Substitution in (63) gives the following certified one-sided bounds:

$$\begin{aligned} a_1(2440^2 - 2\cdot 2440 - 1) &> 390371.6, \\ \frac{\log 2}{4}\big(4878U_{\max} + 2V_{\max} + 4U_{\max}\big) &< 72.1, \\ S^*\big(2D(2440) + D(2441)\big) &< 276163.2, \\ \alpha_+(3\cdot 2440 + 4) &< 112977.6, \\ 3\left(\frac{1.45}{2440} + 10^{-3}\right) &< 0.005. \end{aligned}$$

Therefore

$$\varphi(2440) > 390371.6 - 72.1 - 276163.2 - 112977.6 - 0.005 > 1158.$$

The same substitution gives $\varphi(2432) < 0 < \varphi(2433)$, so the criterion first closes at $k = 2433$, with margin 139. We carry 2440, which (60) covers. □

Lemma 6.3. *The function $k \mapsto \varphi(k)$ is strictly increasing for integers $k \geq 2440$.*

Proof. First,

$$a_1\big((k+1)^2 - 2(k+1) - 1 - (k^2 - 2k - 1)\big) = a_1(2k-1).$$

Moreover

$$\begin{aligned} D(m+1) - D(m) &= \log 2 \left(\log(m+2) + m \log\left(1 + \frac{1}{m+1}\right)\right) \\ &\le \log 2\big(\log(m+2) + 1\big). \end{aligned}$$

The last term in (62) decreases with k. Hence

$$\begin{aligned} R(k+1) - R(k) \le \frac{\log 2}{2} U_{\max} + 3\alpha_+ \\ + S^* \log 2\big(2\log(k+2) + \log(k+3) + 3\big). \end{aligned} \tag{65}$$

Let the right side of

$$\begin{aligned} M(x) = a_1(2x-1) - \frac{\log 2}{2} U_{\max} - 3\alpha_+ \\ - S^* \log 2\big(2\log(x+2) + \log(x+3) + 3\big) \end{aligned}$$

be defined for real $x \ge 2440$. Then

$$M'(x) = 2a_1 - S^* \log 2 \left(\frac{2}{x+2} + \frac{1}{x+3} \right) > 0,$$

because the certified lower bound for d gives $2a_1 > 0.1312$, whereas the subtracted term is < 0.006 for $x \ge 2440$. Direct substitution gives $M(2440) > 146$. Thus $M(k) > 0$ for every $k \ge 2440$. Together with (65), this yields $\varphi(k+1) - \varphi(k) > 0$. □

Proof of Theorem 1.1. By Lemmas 6.2 and 6.3, $\varphi(k) > 0$ for every $k \ge 2440$. Proposition 6.1 therefore gives $f_k < f_{k+1}$ on that range. The certified ranges in (60) cover every integer from 6 through 2439. Together the analytic and certified ranges cover all $k \ge 6$. □

7. A general principle

Conjecture 7.1. Let $\mathcal{P}$ be a set of primes with least element q and thin complement,

$$\sum_{p \notin \mathcal{P}} \frac{1}{p} < \infty, \tag{66}$$

and let $f_k^{\mathcal{P}}$ be the sum of $1/(n \log n)$ over integers n with $\Omega(n) = k$ and all prime factors in $\mathcal{P}$. The sequence $f_k^{\mathcal{P}}$ is eventually increasing if and only if $q = 2$.

Some hypothesis of this kind is needed. For $\mathcal{P} = \{2\}$ one has $f_k^{\mathcal{P}} = 1/(2^k k \log 2)$, and for $\mathcal{P} = \{2, 3\}$ the sequence again decreases to 0; in both cases $q = 2$ and the conclusion fails. What goes wrong is that a sparse $\mathcal{P}$ leaves $\sum_{p \in \mathcal{P}} p^{-s}$ bounded as $s \to 1^+$, so the generating function has no $\zeta(s)^z$ factor and $f_k^{\mathcal{P}}$ decays geometrically to 0 whatever q is.

Under (66) that cannot happen, and the singularity structure is the one the two theorems above exploit:

$$\prod_{p \in \mathcal{P}} (1 - zp^{-s})^{-1} = \prod_{p} (1 - zp^{-s})^{-1} \prod_{p \notin \mathcal{P}} (1 - zp^{-s}),$$

where the second product converges locally uniformly by (66) and is therefore entire in z. Its zeros cancel exactly the poles at $z = p^s$ with $p \notin \mathcal{P}$, leaving the first z-singularity at $z = q^s$. At $s = 1$ the point $z = 2$ lies on that singular circle exactly when $q = 2$. If $q \ge 3$ the coefficient extraction at $z = 2$ is interior and gives a constant multiple of 2^{-k}, as in Section 3; if $q = 2$ it takes place at the boundary and may produce the factor k^2.

Three points remain to turn this mechanism into a theorem for arbitrary $\mathcal{P}$ satisfying (66). First, the regularized product must provide enough uniform control to identify the limiting constant and the coefficient at $z = 2$. Second, in the boundary case one must rule out cancellation of the k^2 coefficient. Third, in the interior case one must show that the constant multiplying 2^{-k} is positive, which is what makes the sequence decrease rather than increase; for $\mathcal{P} = \{p : p \geq 3\}$ this is the inequality $\beta > 0$ of (5). The present paper proves the two extremal cases $\mathcal{P} = \{\text{all primes}\}$ and $\mathcal{P} = \{p : p \geq 3\}$.

8. Certification and formalization

The logical division is as follows.

The analytic proof consists of Lemma 2.1, Proposition 2.2, the coefficient identities (13) and (48), the remainder reduction in Proposition 3.5, and the source estimates in Sections 5–6. These reductions are formalized in Lean 4 without `sorry`.

The numerical propositions are Proposition 3.3 and Proposition 5.6. Their statements are finite systems of interval inequalities. They are established by Arb/FLINT through its Python bindings. The Lean formalization imports the resulting intervals as hypotheses and proves that those hypotheses imply Theorems 1.1 and 1.2.

Two constants are stated above more conservatively than the computation requires, so that the text and the formalization carry the same values: the threshold 2440 of Lemma 6.2 in place of 2433, and (61) in place of the sharper bound that Lemma 3.4(i) supplies. Neither changes the ranges covered.

For the odd theorem, the coefficient bounds in (21) are obtained by cancelling the pole at $z = 3$ before applying coefficient recursion or Cauchy bounds. The range $0 < u \leq 3$ is evaluated directly, while the range $u \geq 3$ uses the Taylor model made explicit in Proposition 3.1.

The constant C_E is certified on the whole of $(1, s(3)]$. The four quantities entering it are ratios in which the pole of ζ at $s = 1$ cancels. Interval evaluation on boxes of s loses that cancellation, since each occurrence of $s - 1$ is widened independently, and the enclosures diverge however finely $(1, s(3)]$ is partitioned. Expanding ζ as a power series in $\sigma = s - s_0$ about $s_0 = 1 + E/2$, where $E = s(3) - 1$, makes the cancellation symbolic. The resulting series are evaluated on $|\sigma| \leq E/2$, which is the whole range; their coefficients fall off geometrically, and truncating after nine terms contributes less than 10^{-20}.

The constant C_N requires $\Pi(t) = \sum_{p \geq 3} p^{-t}$ at $t = s, 2s, \ldots, 200s$ for $s = s(3)$. At $s \approx 1.05$ these converge too slowly to truncate: the Rosser–Schoenfeld bound applied beyond 10^8 still overestimates the tail twofold, and that excess compounds through the coefficient recursion. Instead Π is evaluated by the Möbius series $\Pi(t) = \sum_{n \geq 1} \mu(n) n^{-1} \log \zeta(nt) - 2^{-t}$, which requires no prime enumeration. The recursion itself is run at 600 bits, since cancellation costs about one digit every 2.3 steps over the 200 terms used; at 1200 bits the head agrees to every printed digit.

For the unrestricted theorem, the s-uniform coefficient bounds defining α_+ are obtained by carrying s as an interval over a partition of $[1, 2]$. The signed moments U and V are evaluated before absolute values are taken. This order is essential: replacing them by $\sum_i 2^i |c_i|$ loses the cancellation used in (53).

Appendix A. Certified quadrature

The direct values of f_k are computed from the transformed integrand $F(u) = P_k(1 + \mathrm{e}^{-u})\mathrm{e}^{-u}$, which is analytic on the strip $|\operatorname{Im} u| < \pi/2$. The range $u \in [-5.10,\, 3.5k]$ is divided into panels of width $H = 0.8$, each mapped affinely to $[-1, 1]$ and integrated by Gauss–Legendre quadrature

with $N = 52$ nodes at 1400 bits of precision; the number of panels is therefore proportional to k.

For f analytic inside the Bernstein ellipse E_ρ and bounded there by M, the N-node Gauss rule satisfies

$$\left| \int_{-1}^{1} f(x)\,\mathrm{d}x - \widehat{I}_N \right| \le \frac{64}{15}\,\frac{M\rho^{-2N}}{1-\rho^{-2}} \tag{67}$$

[5, Thm. 4.5], where the rule is indexed so that I_n uses $n+1$ nodes and the stated exponent $\rho^{-(2n+2)}$ is the ρ^{-2N} above; the bound appears as [6, Thm. 19.3]. The implementation evaluates the right side of (67) with the interval-scaling factor $H/2$ and a further factor 10 of margin. The ellipse is taken with semi-minor axis $\delta = 1.45 < \pi/2$ in the u variable, giving $\rho \approx 7.385$, and M is bounded using $|P_k(s)| \le P_k(\operatorname{Re} s)$ together with the monotonicity of P_k on the reals. The resulting discretization error is below 10^{-88} per panel.

The pinned coefficient sweeps use primes up to 10^5 and 300 bits of precision, with 1600 boxes in s for α_+ and 800 boxes for the second-derivative sweep. The certificate records these parameters together with a SHA-256 hash of the numerical payload.

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